Parallelograms and rectangles

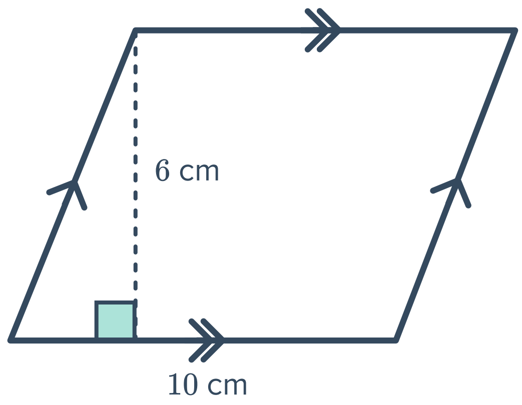
1 For the following parallelogram:

a If the parallelogram is formed into a rectangle, find:

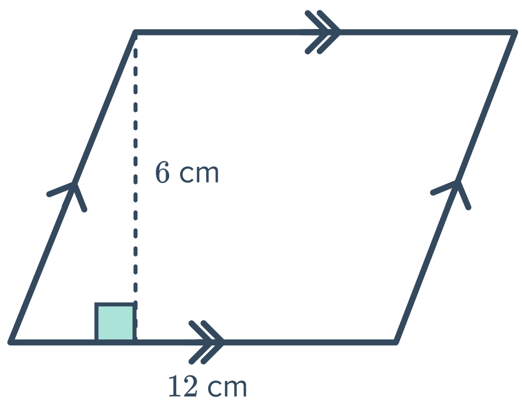
i The length of the rectangle.

ii The width of the rectangle.

b Find the area of the parallelogram.



2 Find the area of the following parallelogram by rearranging it into a rectangle:

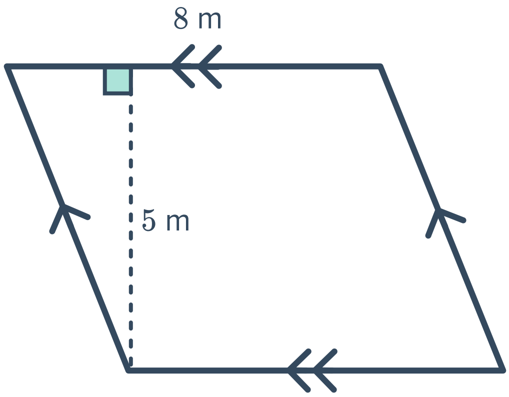


Area of a parallelogram

3 Consider the following parallelogram:

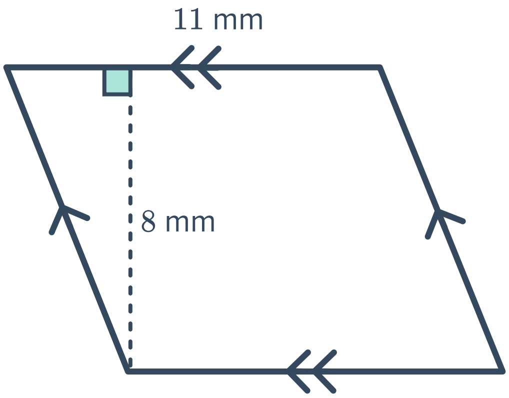
a Identify the base and the height.

b Find the area of the given parallelogram.

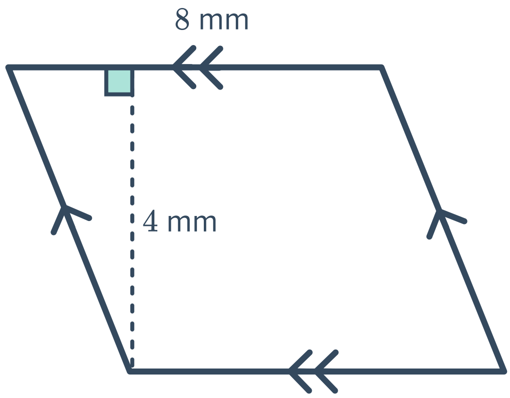


4 Find the area of the following parallelograms:

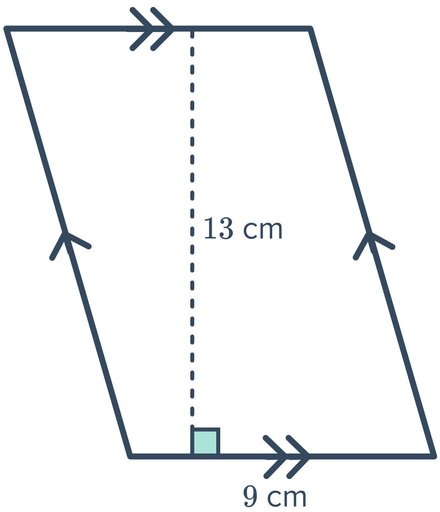
a



b



c



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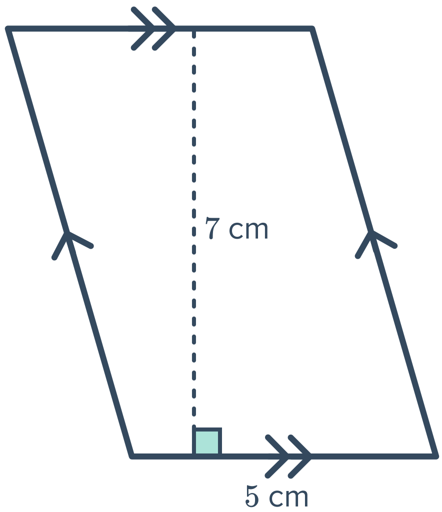

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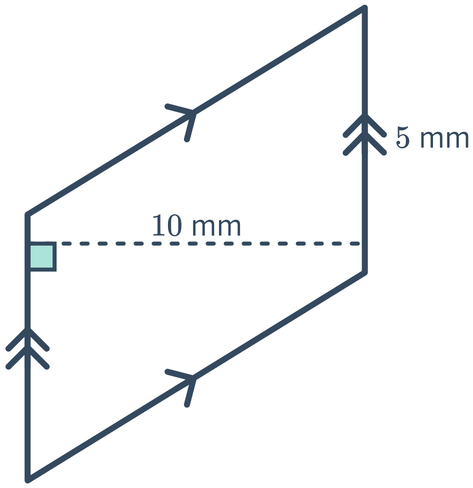

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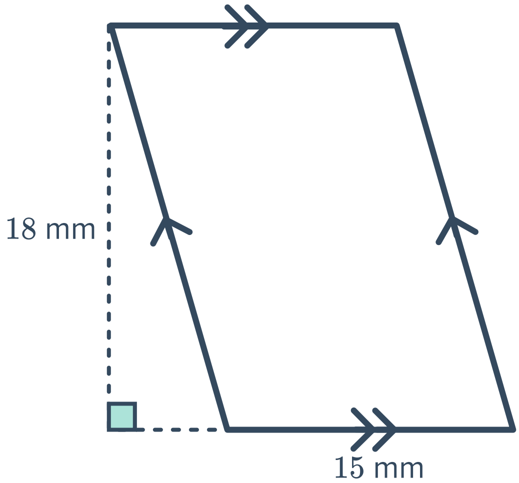

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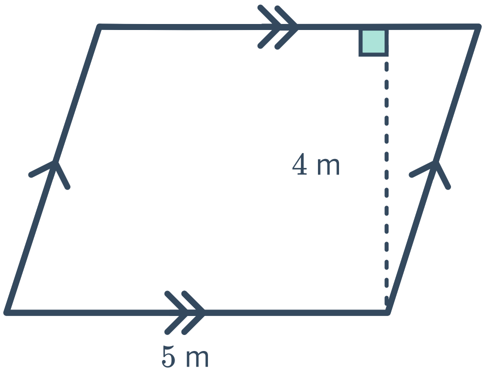

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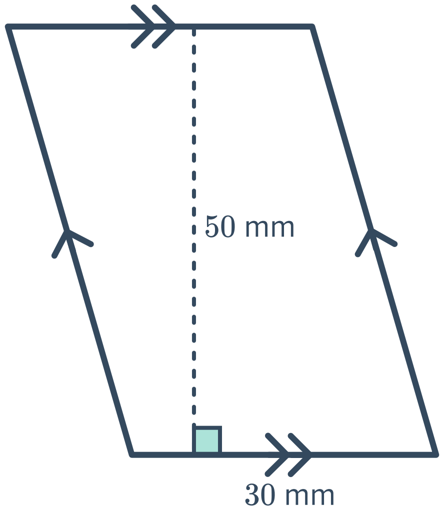

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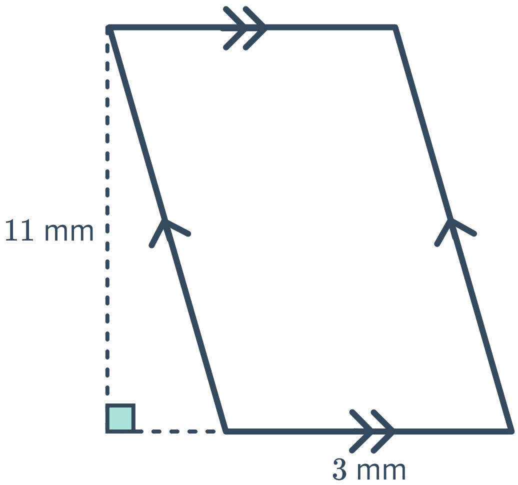

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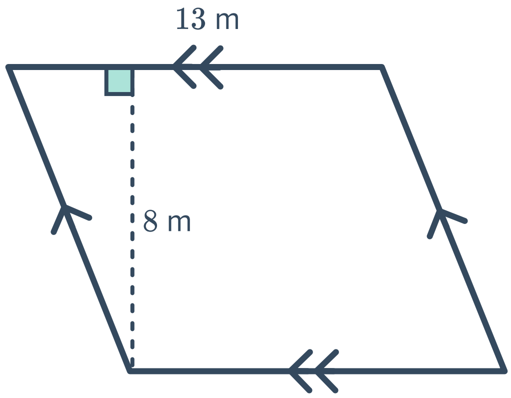

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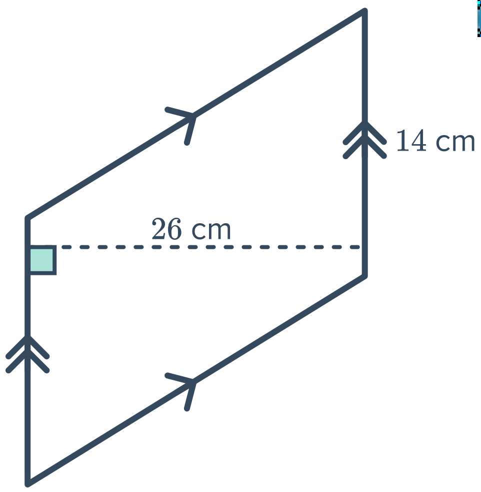

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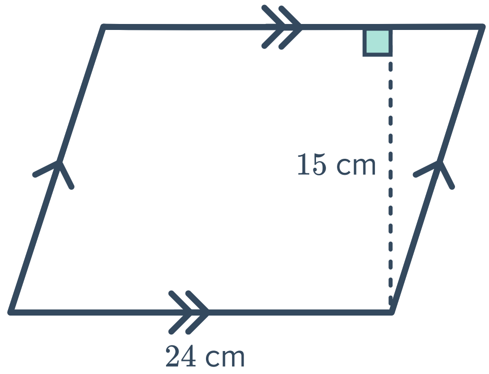
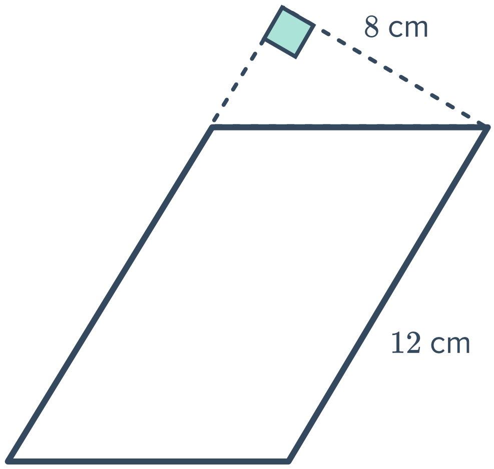


m



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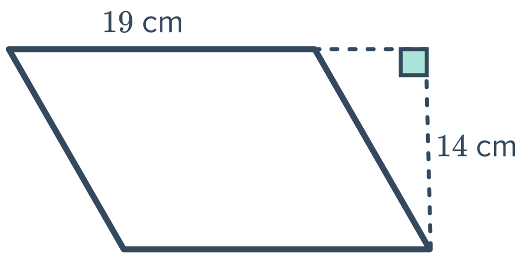


5 Find the area of the parallelograms with the following dimensions:

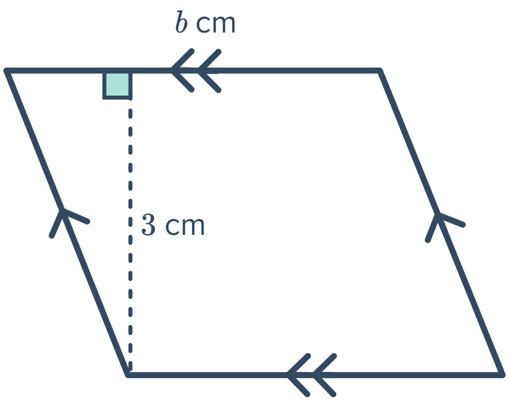
a The base is $5\ \mathrm{mm}$, and the perpendicular height is $2\ \mathrm{mm}$.

b The base is $15\ \mathrm{cm}$ and the perpendicular height is $7\ \mathrm{cm}$.

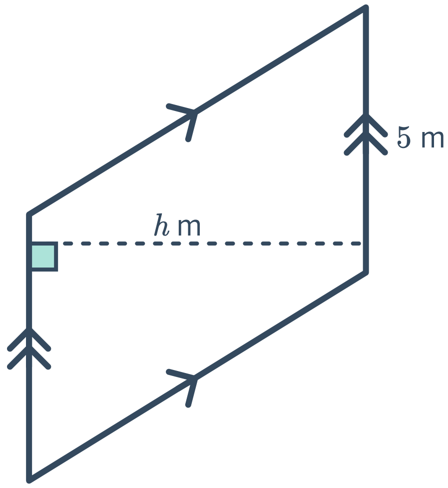
Dimensions from area

6 Find the value of b or h in the following parallelograms:

a Area $= 36 \text{ cm}^2$



b Area $= 35 \text{ m}^2$



7 Find the base length of a parallelogram whose area is 108 cm^2 and perpendicular height is 9 cm.

8 Find the perpendicular height of a parallelogram whose area is 99 m^2 and base is 9 m.

9 Determine whether the following could be the dimensions of a parallelogram with an area of 28 mm^2:

a Base $= 1 \text{ mm}$, height $= 28 \text{ mm}$

b Base $= 7 \text{ mm}$, height $= 4 \text{ mm}$

c Base $= 4 \text{ mm}$, height $= 7 \text{ mm}$

d Base $= 2 \text{ mm}$, height $= 28 \text{ mm}$

10 Complete the following table of base and height measurements for different parallelograms with an area of 60 cm^2:

Base (cm)		10	30
Height (cm)	12		
Area (cm^2)	60	60	60

Applications

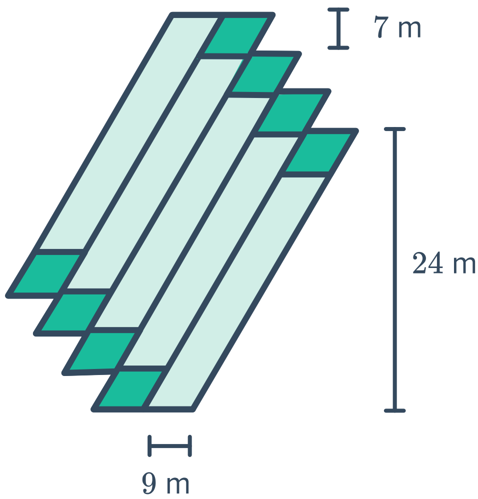
11 Some car parks require the cars to park at an angle as shown. The dimensions of the car park are as given, where each individual parking space has a length of 5.1 m and a width of 3.9 m. What area is needed to create an angled car park suitable for 7 cars?

12 The tile pattern shown is made up of two different parallelograms in a tessellated pattern.

a Find the area covered by the smaller tiles.

b Find the area covered by the larger tiles.

c Calculate the total area covered.



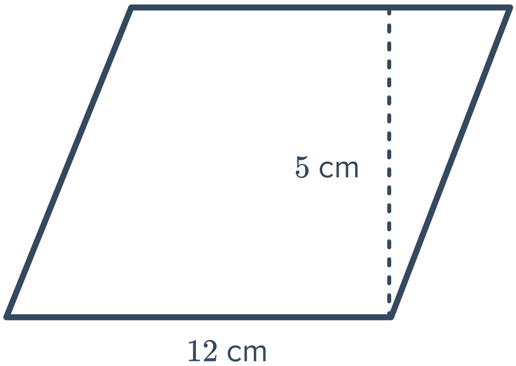
13 James used some scrap paper to make a birthday card in the shape of a parallelogram. The base of the card is 17 cm long, and the perpendicular height is 16 cm. Find the area of the card.

14 A roof in the shape of a parallelogram is to [redacted]ntirely covered by identical solar panels that are also in the shape of a parallelogram. Each solar panel measures $1.2\ \mathrm{m}$ along the longest side and has a perpendicular height of $0.6\ \mathrm{m}$.

a Find the area covered by one solar panel.

b Find the number of solar panels to be installed if the roof measures $23.04\ \mathrm{m}^2$ in area, and no part of the roof is to be left uncovered by solar panels.

15 An area measuring $2280\ \mathrm{cm}^2$ is to be paved with identical tiles in the shape of parallelograms. Each tile measures $12\ \mathrm{cm}$ along the base, and has a perpendicular height of $5\ \mathrm{cm}$.



a Find the area that each tile covers.

b How many tiles are needed to cover the whole area?

16 A quilt is made by sewing together 4 identical parallelograms as shown in the following figure:

If the total area of the quilt is $1944\ \mathrm{cm}^2$, calculate the perpendicular height of each parallelogram piece.